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Tsi12P

Tenkasi District

Common Half Yearly Examination - December 2025

19-12-25

Standard 12

PHYSICS

PART - I

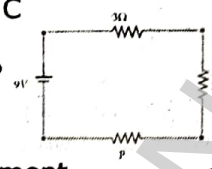
Time Allowed: 3.00 Hours | Maximum Marks: 70

15×1=15

Note: i) Answer all the questions.
ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.

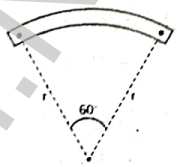
1) A parallel plate capacitor is made by stacking n equally spaced plates connected alternatively. If the capacitance between any two adjacent plates is C, then the resultant capacitance is
a) (n−1)C b) (n+1)C c) C d) nC

2) There is a current of 1.0A in the circuit shown below. What is the resistance of P?



a) 1.5Ω b) 2.5Ω
c) 3.5Ω d) 4.5Ω

3) A bar magnet of length ℓ and magnetic moment Pm is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be _____



a) Pm b) $\frac{3}{\pi}$ Pm c) $\frac{2}{\pi}$ Pm d) $\frac{1}{2}$ Pm

4) For a symmetrical sinusoidal current rms value of current is _____ of its peak value.
a) 63.7% b) 70.7% c) 100% d) 14.14%

5) Consider an oscillator which has a charged particle oscillating about its meanposition with a frequency of 300 Mhz. The wavelength of electromagnetic wave produced by the oscillator is _____
a) 1m b) 10 m c) 100 m d) 1000 m

6) A ray of light travelling in a transparent medium of refractive index n falls on a surface seperating the medium from air at an angle of incidence of 45°. The ray can undergo total internal reflection for the following n
a) n = 1.25 b) n = 1.33 c) n = 1.4 d) n = 1.5

7) In a Young's double slit experiment the slit seperation is doubled. To maintain the same fringe spacing on the screen, the screen to slit distance D must be changed to _____
a) 2D b) $\frac{D}{2}$ c) $\sqrt{2}$ D d) $\frac{D}{\sqrt{2}}$

8) A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\lambda/2$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case the workfunction at the surface of the material is _____
a) $\frac{hc}{\lambda}$ b) $\frac{2hc}{\lambda}$ c) $\frac{hc}{3\lambda}$ d) $\frac{hc}{2\lambda}$

9) Atomic number of H-like atom with ionization potential 122.4V for n = 1 is
a) 1 b) 2 c) 3 d) 4

10) The barrier potential of a silicon diode is approximately
a) 0.7 V b) 0.3 V c) 2.0 V d) 2.2 V

11) Which one of the following is a natural nanomaterial
a) Peacock feather b) Peacock beak
c) Grain of Sand d) Skin of the Whale

12) If the input to the NOT gate is A = 1011, its output is _____
a) 0100 b) 1000 c) 1100 d) 0011

13) The halflife of radioactive element is 4 days. The time required to disintegrate $\frac{3}{4}^{th}$ of that element is _____
a) 3 days b) 4 days c) 8 days d) 12 days

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14) At curie temperature, ferromagnetic material becomes
a) diamagnetic material b) paramagnetic material
c) antiferro magnetic material d) non magnetic material

15) An electric flux through any one of the sides of cube when an electric charge q is placed at center of the cube is _____

a) $\frac{2\pi q}{6(4\pi \epsilon_0)}$ b) $\frac{4\pi q}{6(4\pi \epsilon_0)}$ c) $\frac{\pi q}{6(4\pi \epsilon_0)}$ d) $\frac{q}{6(4\pi \epsilon_0)}$

PART - II

6×2=12

Note: i) Answer any six quesitons. ii) Q.No. 24 is compulsory.

16) What is meant by quantization of charges?
17) Distinguish between drift velocity and mobility.
18) State Fleming's left hand rule.
19) Why are e.m. waves non-mechanical?
20) What are the two conditions to occur total internal reflection?
21) Define workfunction of a metal. Give its unit.
22) Write down the properties of neutrino.
23) List the applications of light emitting diode.
24) A spherical stone and a spherical metallic shell of same size and mass are dropped from the same height. Which one, a stone or a metal ball will reach the Earth's surface first? Justify your answer. Assume that there is no friction.

PART - III

6×3=18

Note: i) Answer any six questions. ii) Q.No. 33 is compulsory.

25) State and Prove De Morgan's first and second theorem.
26) Derive an expression for deBroglie wavelength of electrons.
27) Two light sources of equal amplitudes interfere with each other. Calculate the ratio of maximum and minimum intensities.
28) Discuss the conversion of galvanometer into ammeter.
29) Mention the various energy losses in a transformer.
30) Derive the expression for equivalent capacitance when capacitors are connected in parallel.
31) List out three properties of nuclear force.
32) Write down Maxwell's equations in integral form.
33) An object of 4 cm height is placed at 6 cm in front of a concave mirror of radius of curvature 24 cm. Find the position, height, magnification and nature of the image.

PART - IV

5×5=25

Note: Answer all the questions.

34) a] Calculate the electric field due to an electric dipole on its axial line.

(OR)

b] Describe the Fizeau's method to determine the speed of light.

35) a] Explain the determination of the internal resistance of a cell using voltmeter.

(OR)

b] Discuss the diffraction at single slit and obtain the condition for n^{th} minimum.

36) a] Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law.

(OR)

b] List out the laws of photoelectric effect.

37) a] Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.

(OR)

b] Explain the J.J.Thomson experiment to determine the specific charge of the electron.

38) a] What is emission spectrum? Explain the types of emission spectrum.

(OR)

b] Draw the circuit diagram of a half wave rectifier and explain its working.

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